

Experiment no-11

11. Implement a simple top-down parser for context-free grammars using python.

PROGRAM:

```
import nltk
from nltk import CFG
from nltk.parse import RecursiveDescentParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Pronoun | Det Adj N | Det N
VP -> V NP | V Adj

Pronoun -> 'he'
Det -> 'a' | 'the'
Adj -> 'good'
N -> 'man'
V -> 'is'
""")

parser = RecursiveDescentParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

for tree in parser.parse(sentence):
    print(tree)
```

OUTPUT:

```
... Enter a sentence: he is a good man
(S (NP (Pronoun he)) (VP (V is) (NP (Det a) (Adj good) (N man))))
```